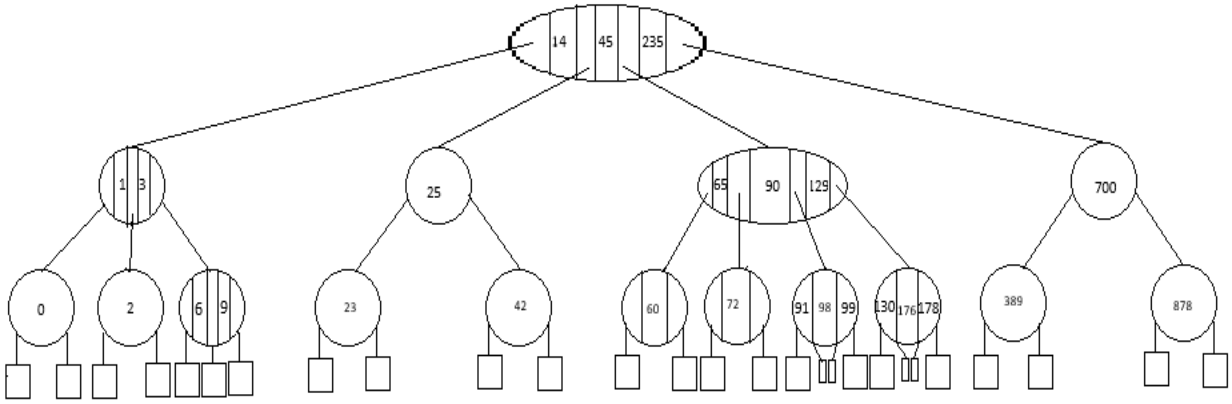
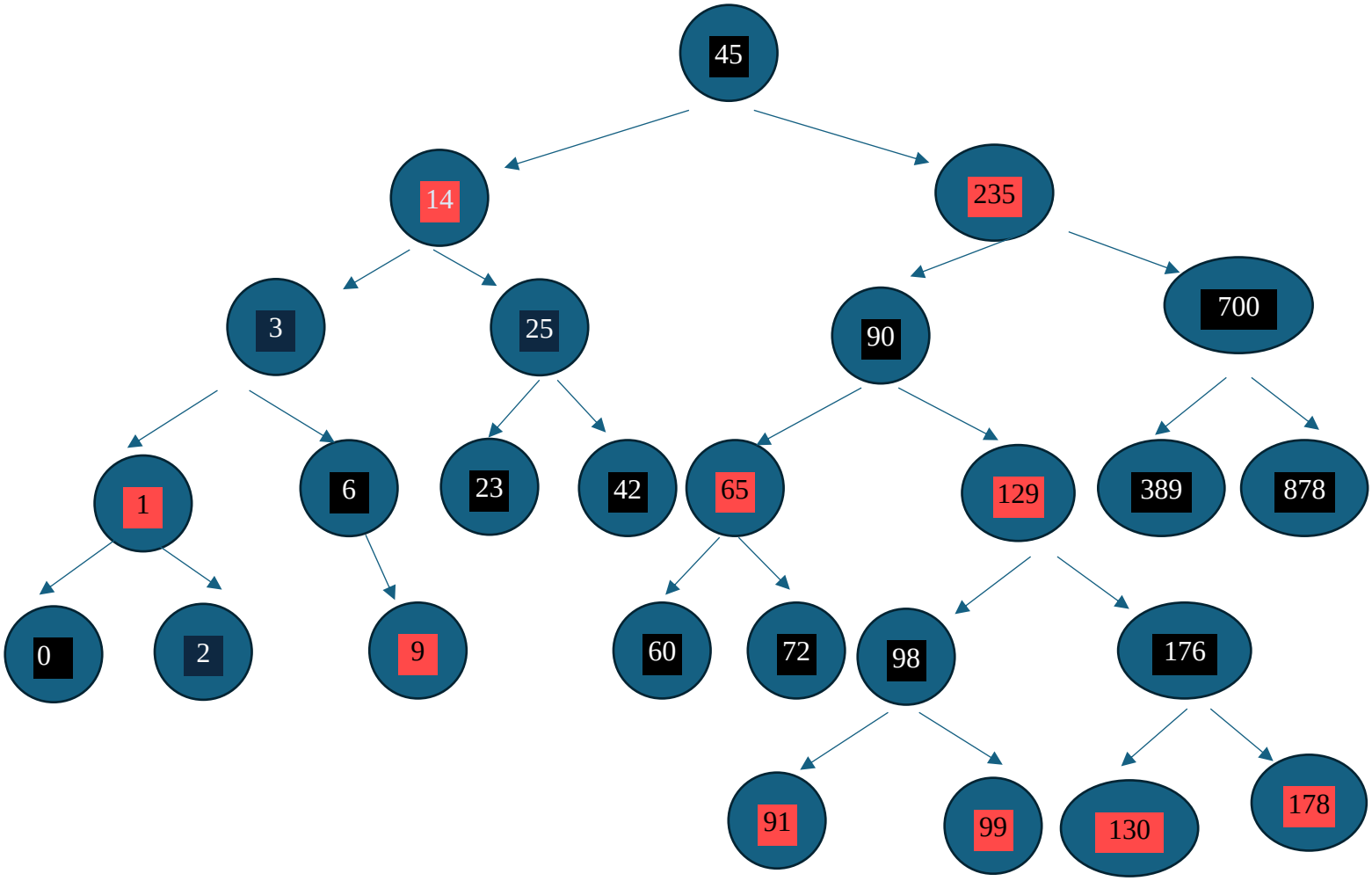
Homework 6

Assigned: October 15, 2024 (Wednesday)
Due: October 22, 2024 (Wednesday)

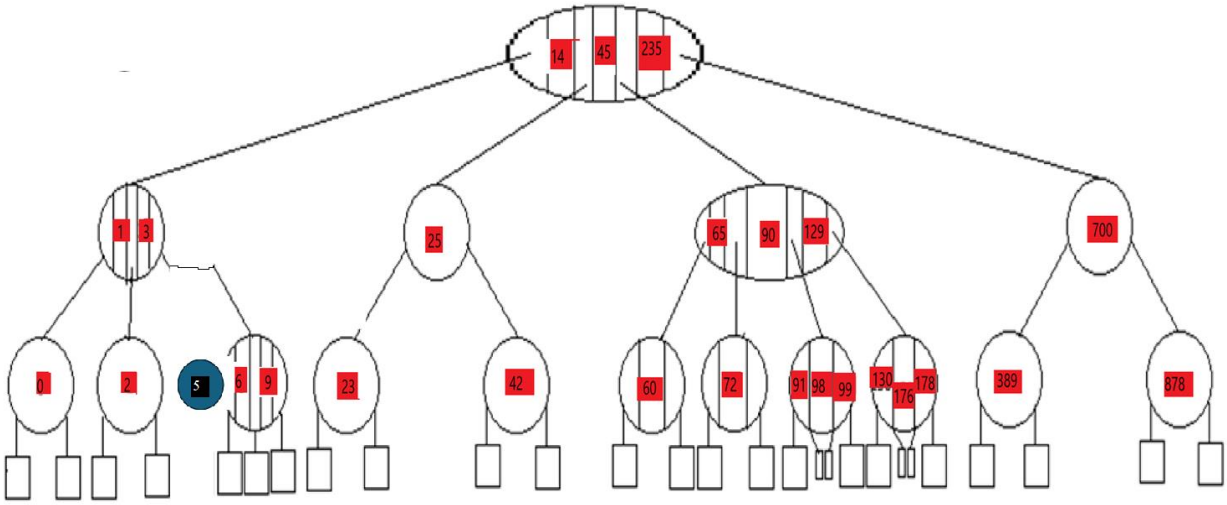
Problem 1:
Convert the following 2-4 Tree into Red Black Tree. Depict the red nodes and edges with double circles and double line segments. [4-Marks]

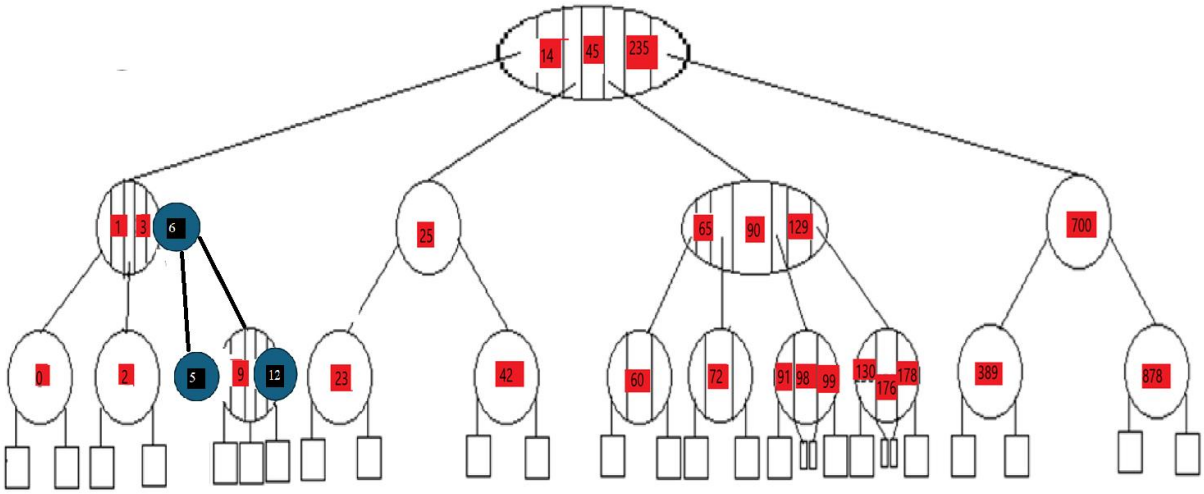


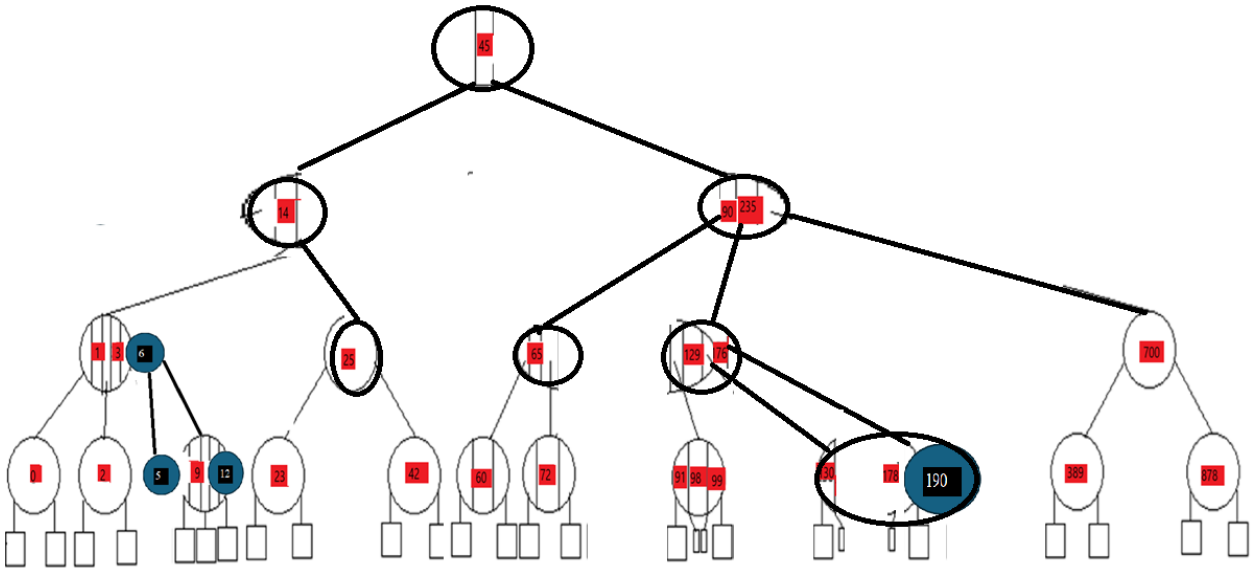


Problem 2: [4 Marks]

a. **Insert 5, 12, 190 to 2-4-tree and the RB-tree obtained in Question 1.**



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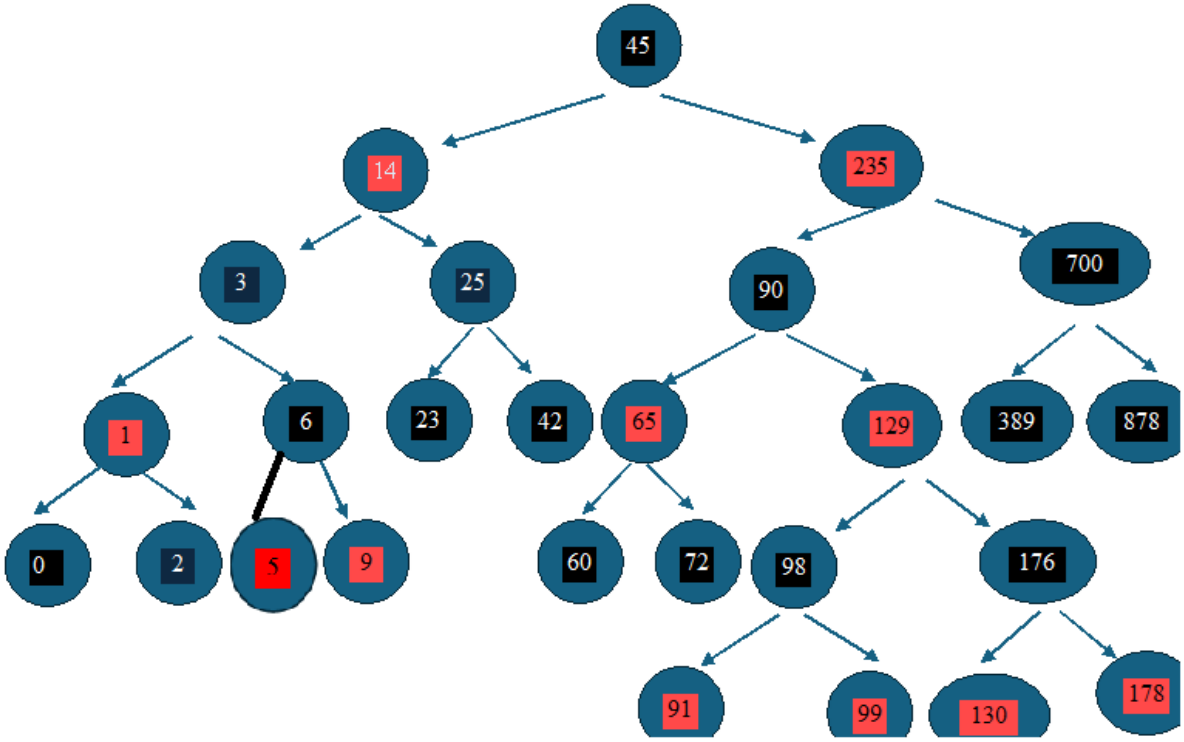
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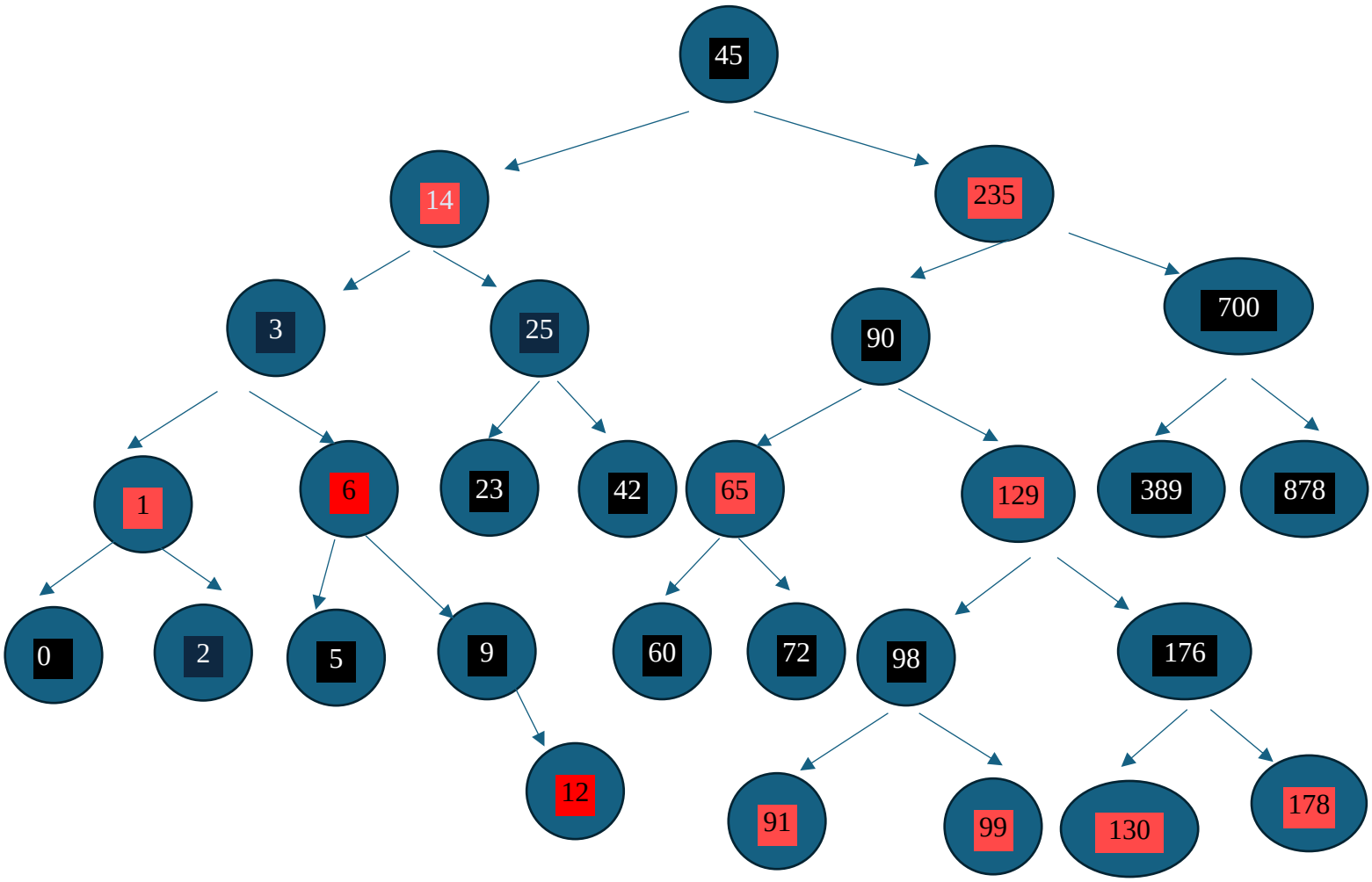
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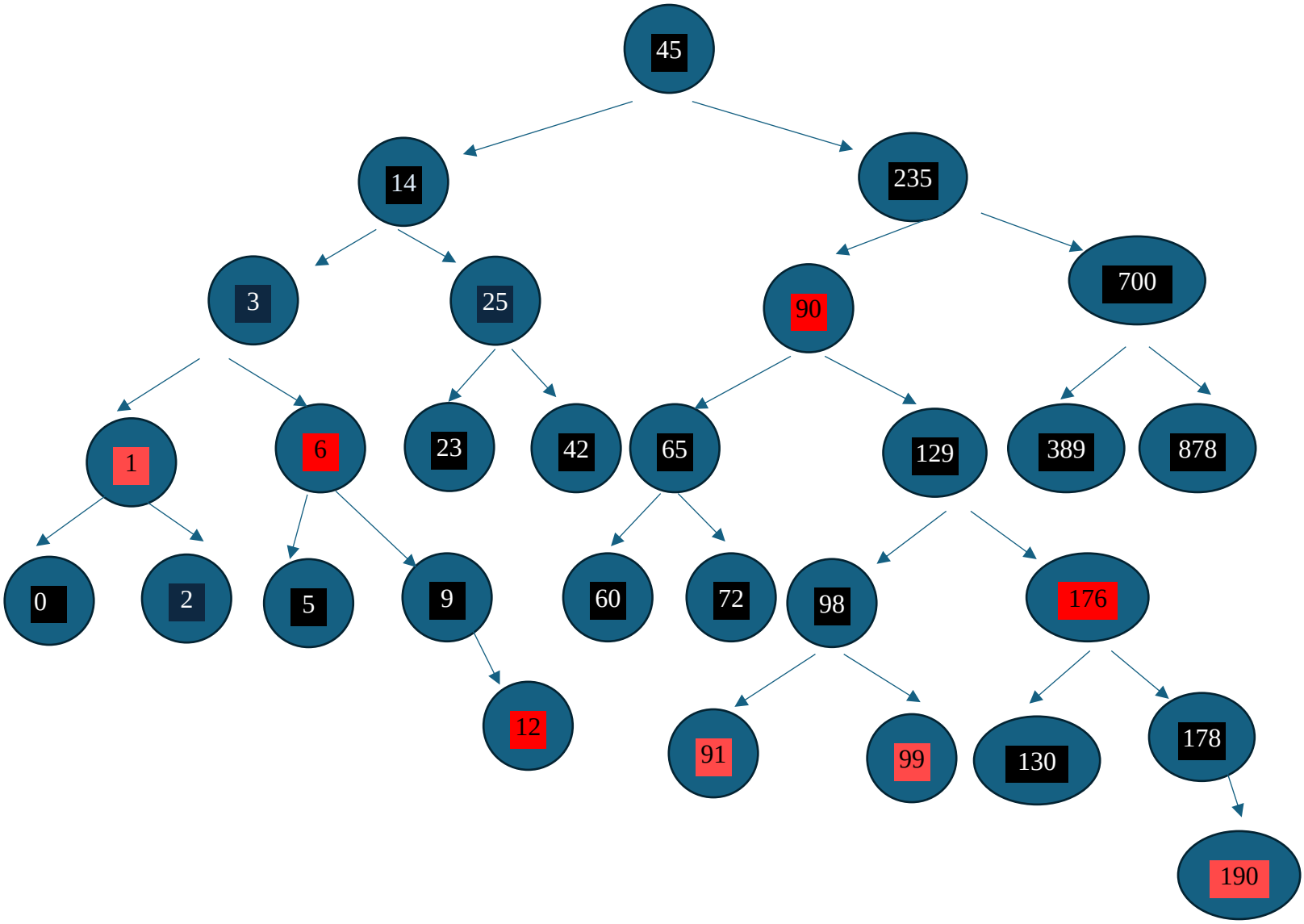
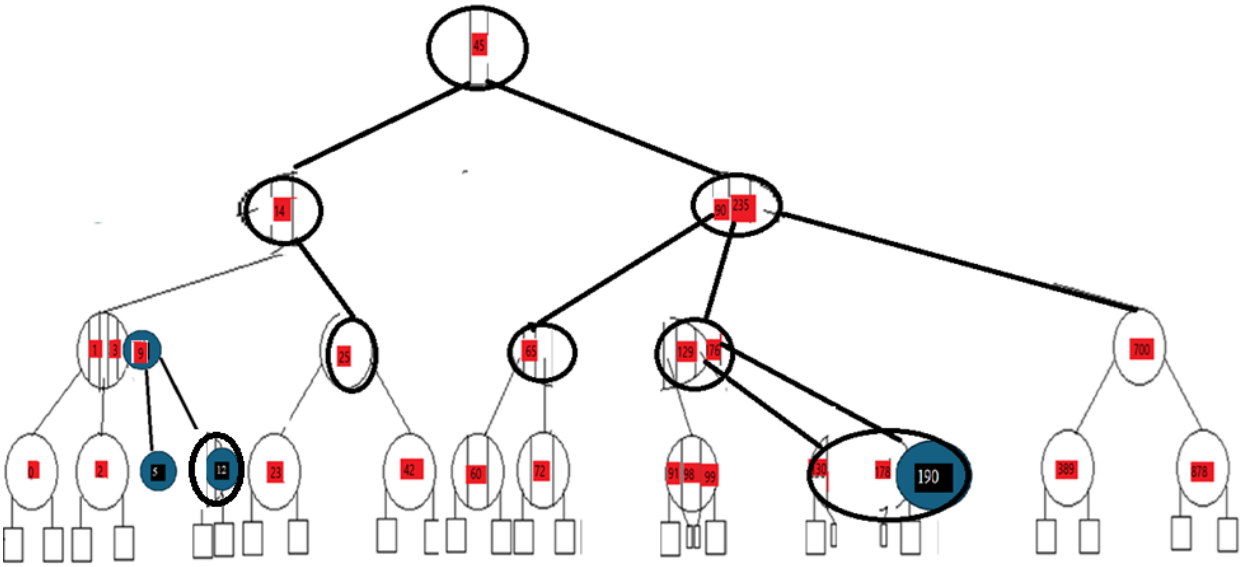
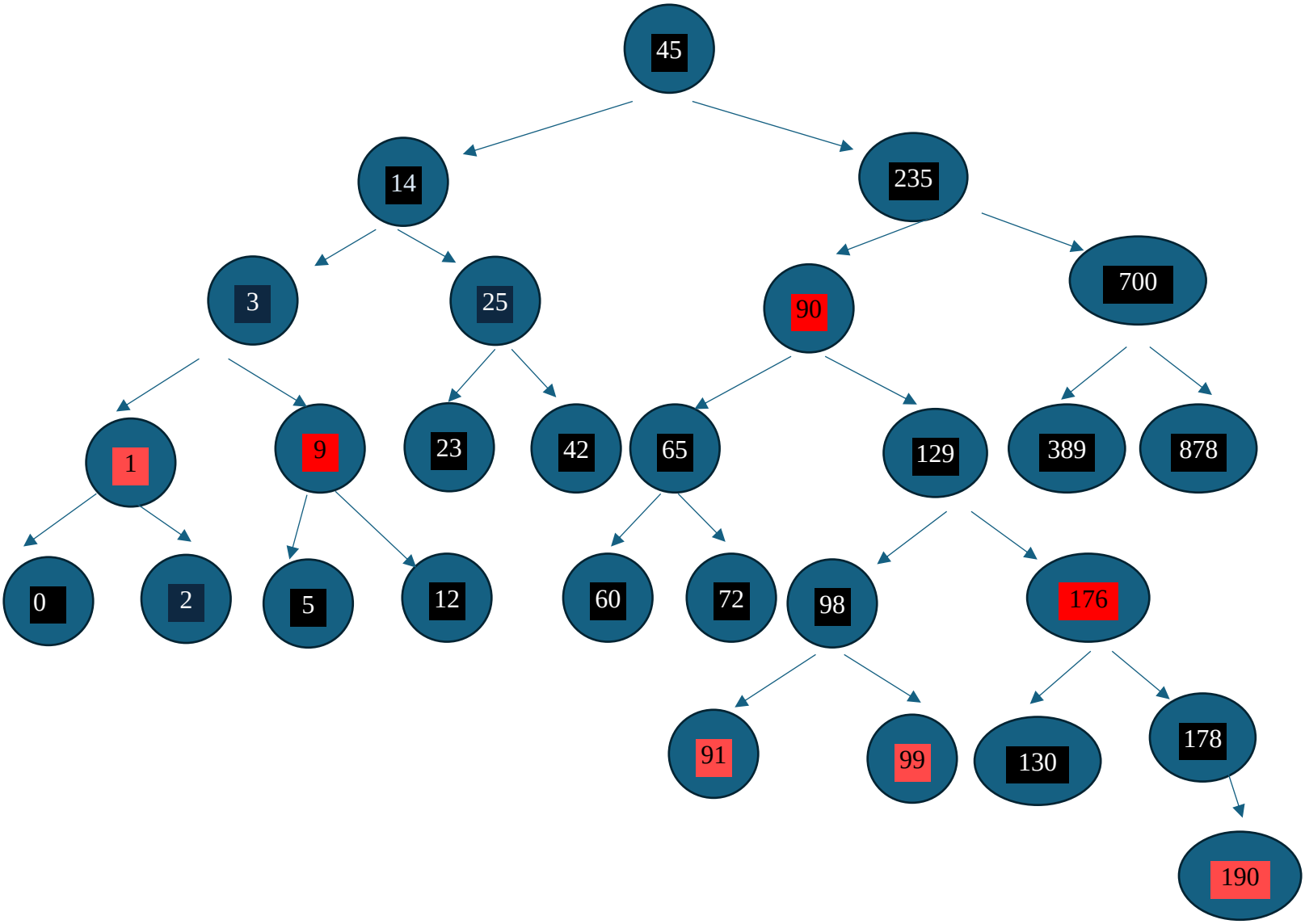


b. Delete 6, 0, 45 and 878 from 2-4-tree and RB-tree after insertions in problem 2a.

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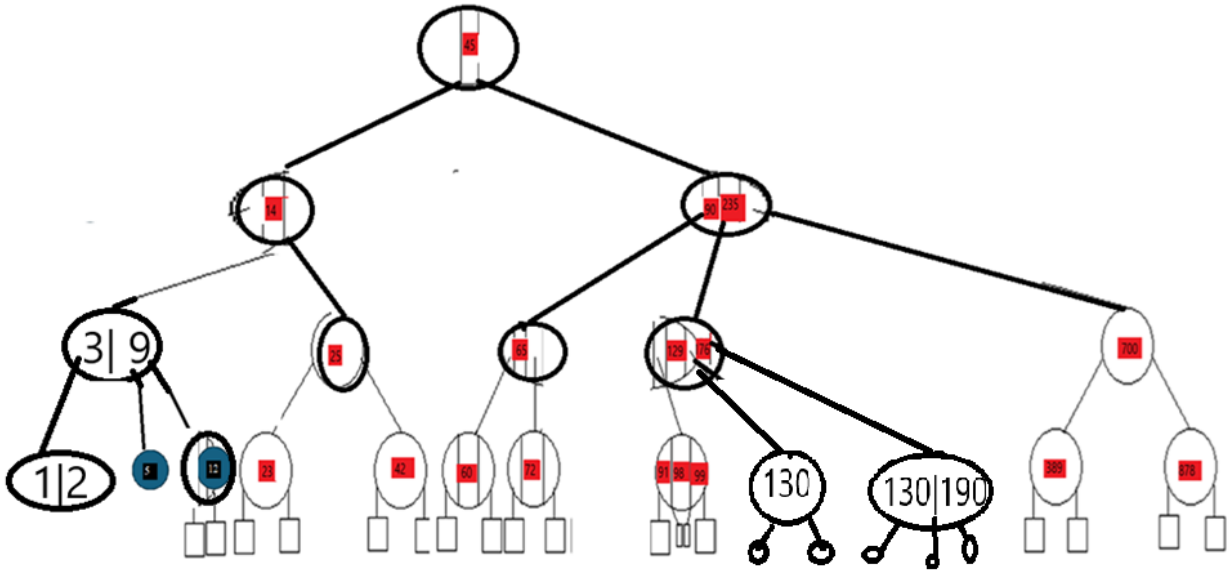
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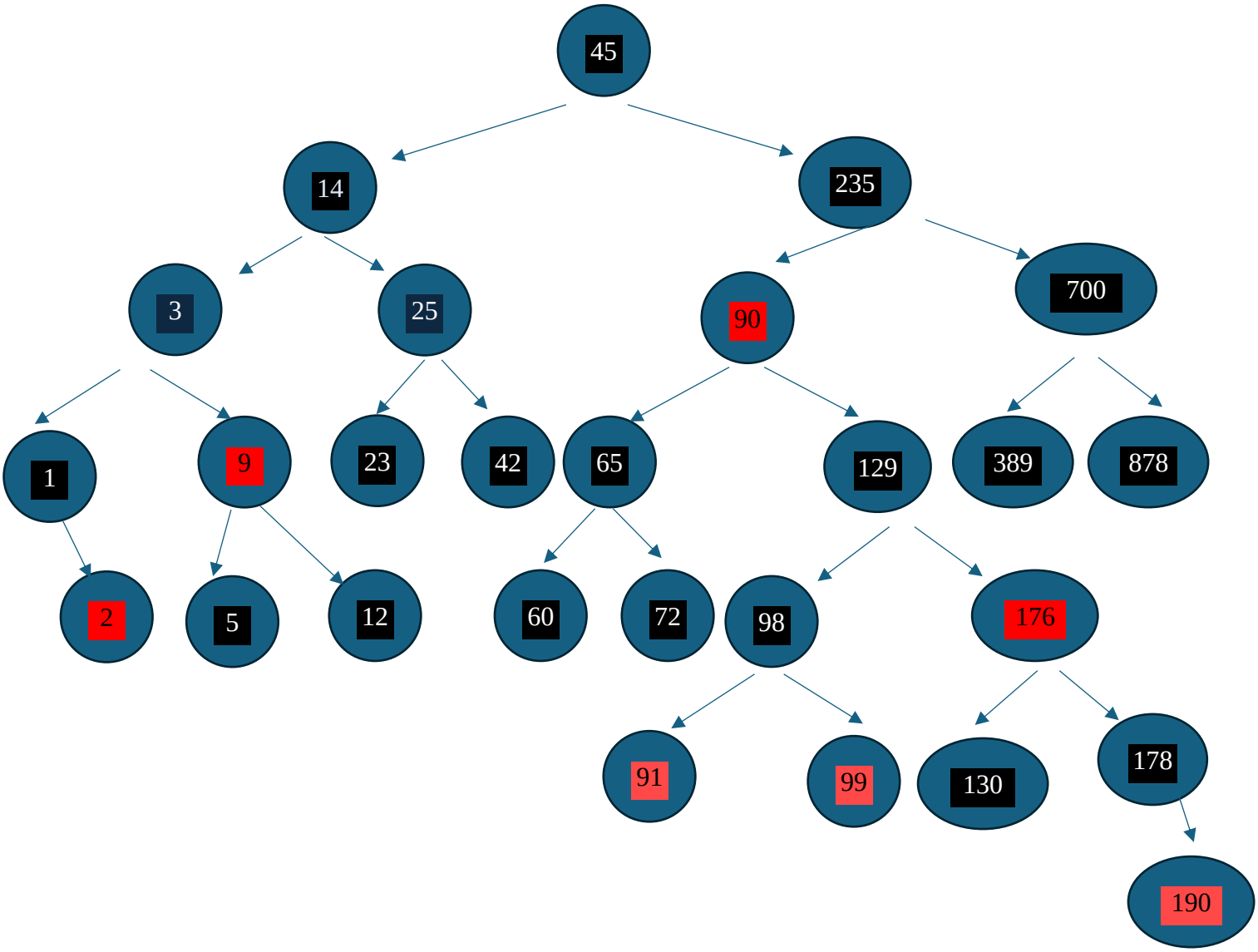
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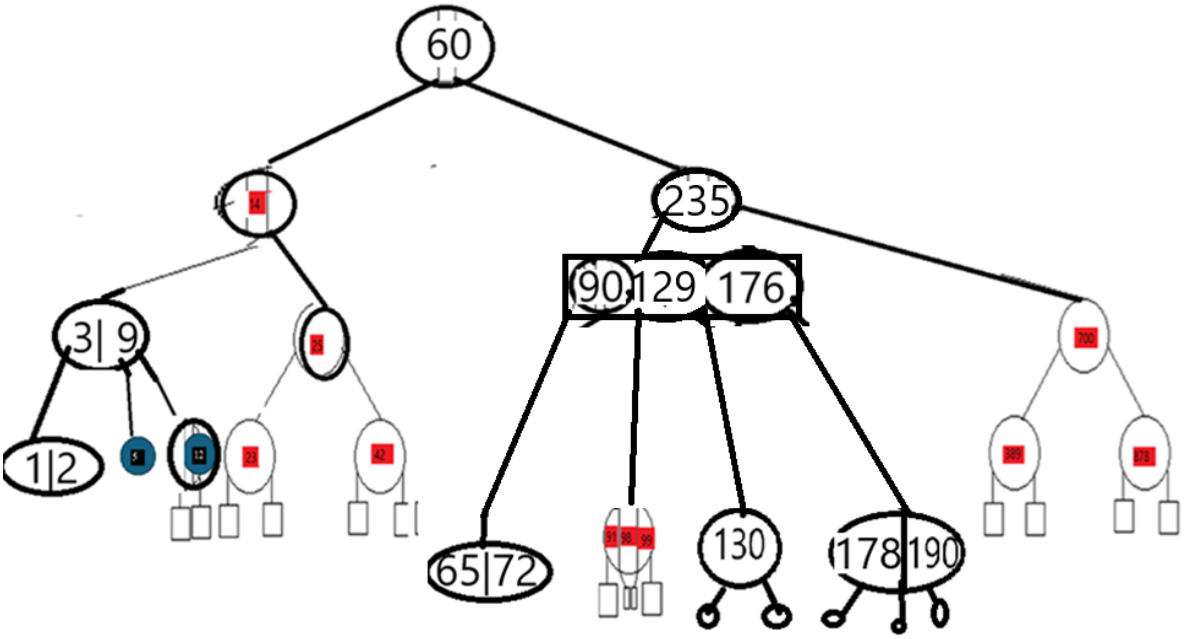
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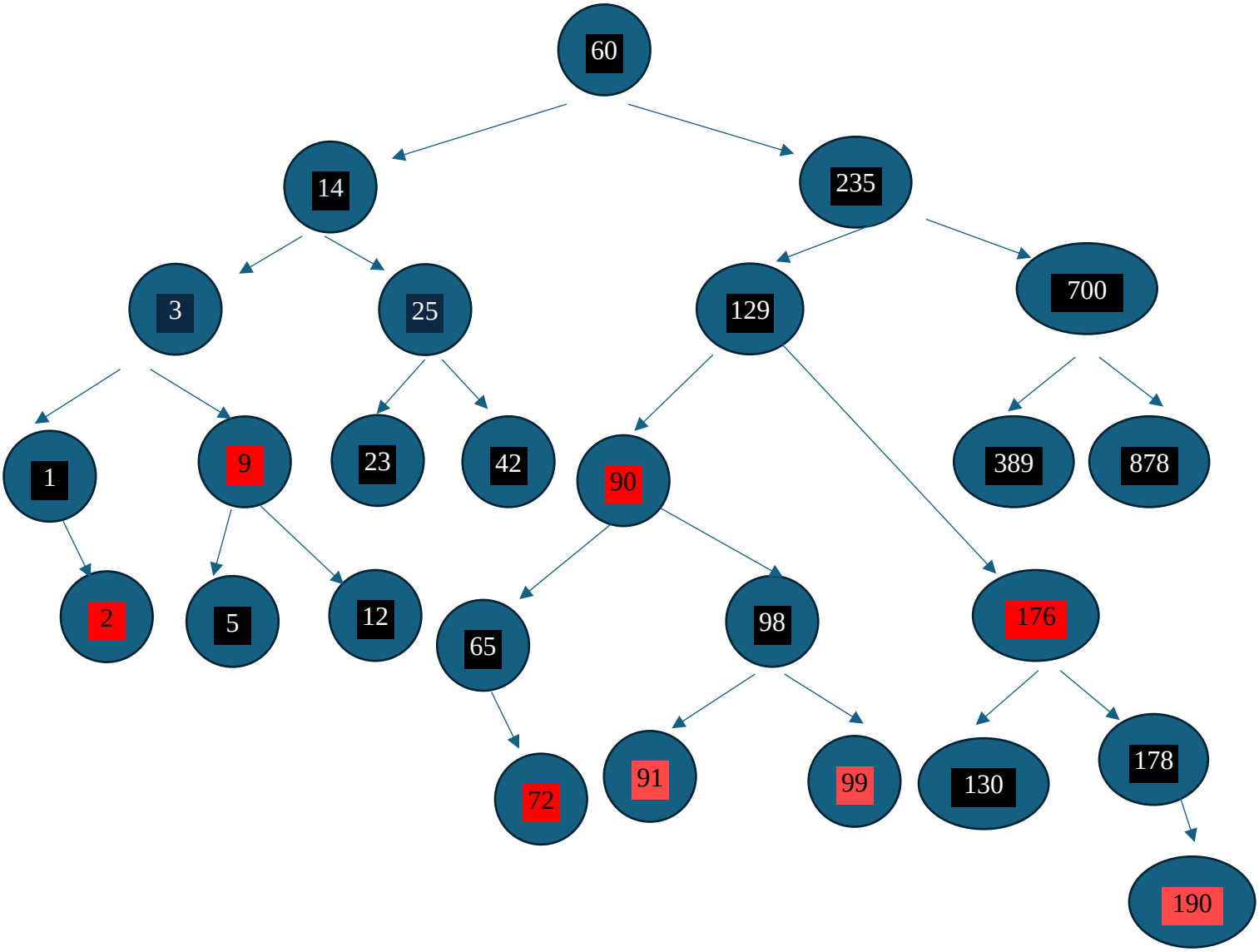
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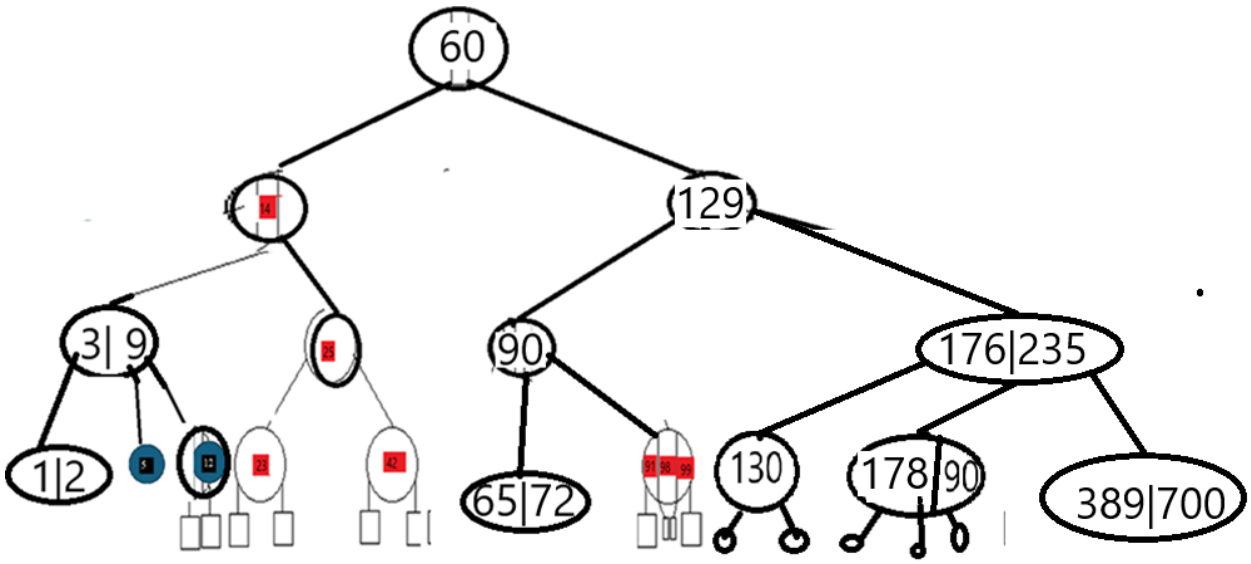
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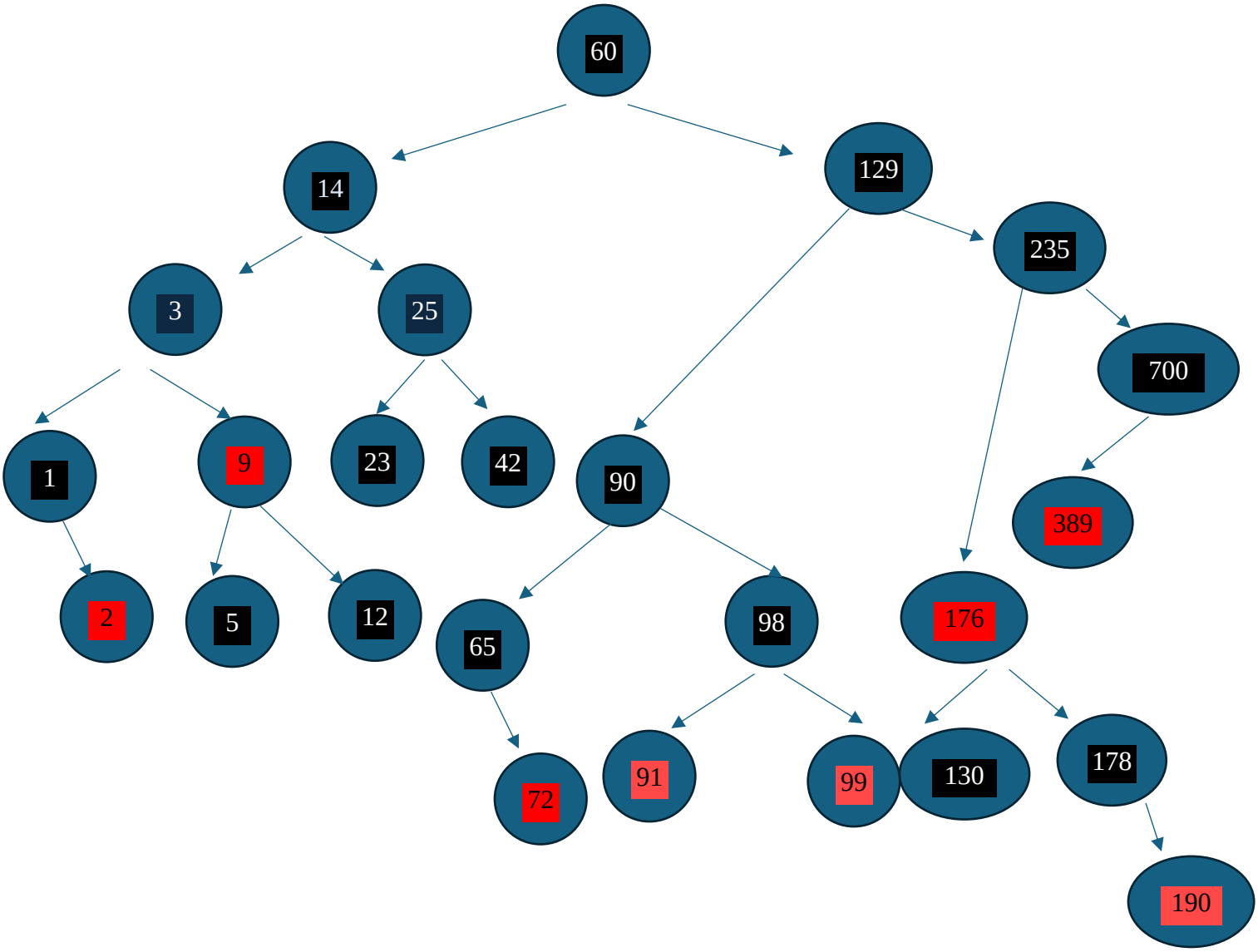
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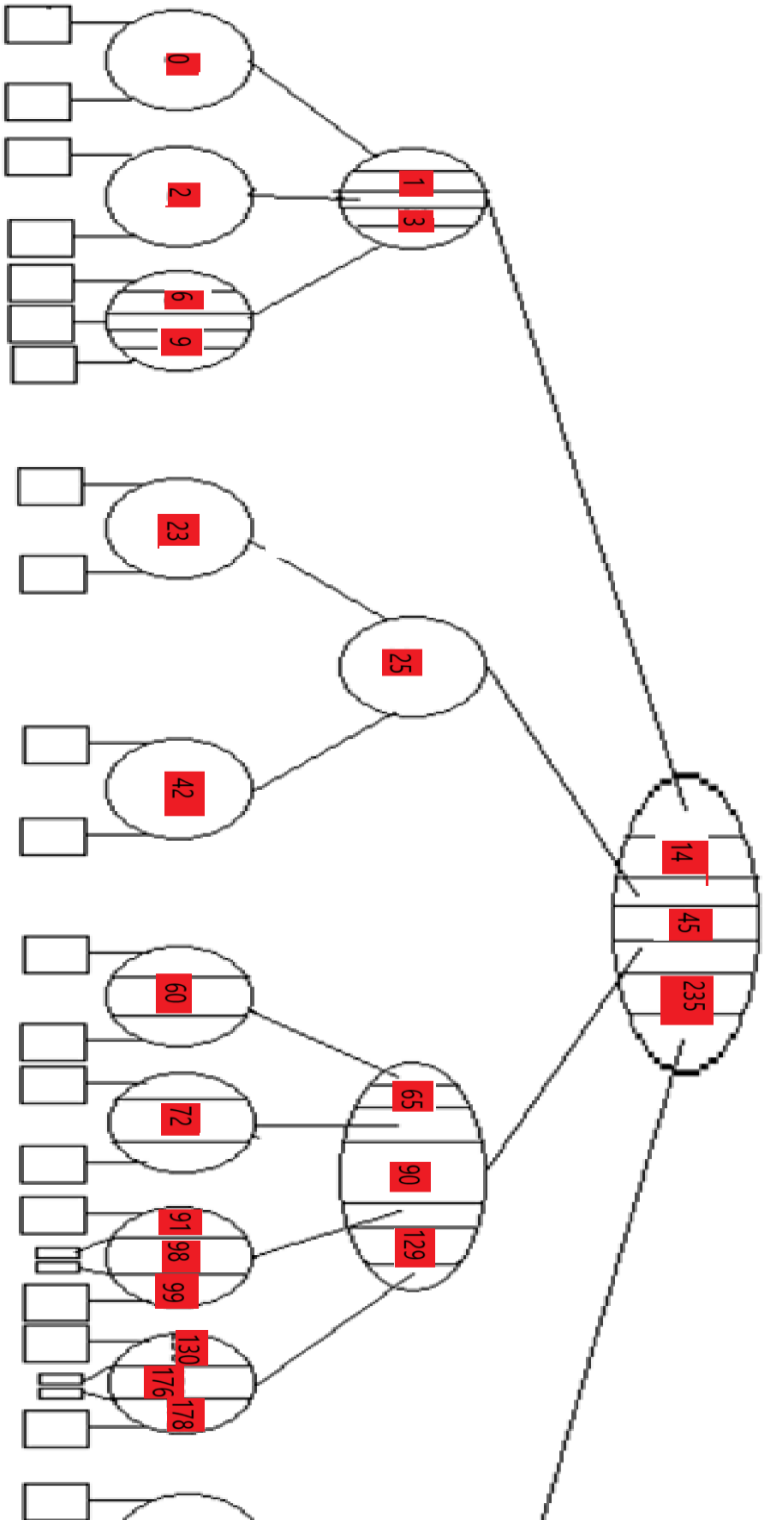
Problem 3: A binary tree with 4000 elements has a height of "h". Find the minimum value of "h" for following trees. [2 Marks]

a. Binary Search Tree h=ceiling(log_2(4000+1)) = 12
b. AVL-Tree h=ceiling(log_2(4000+1)) = 12
c. 2-4-Tree h=ceiling(log_4(4000+1)) = 6
d. Red-Black Tree h=ceiling(log_2(4000+1)) = 12

Problem 4: Find the maximum and minimum height of the following trees with 8,5 and 12 elements. [2 Marks]

a. Binary Search Tree h_min=ceiling(log_2(8+1)) = 4,h_min(5) = 3,h_min(12)=4
    a. H_max = 8,5,12
b. AVL-Tree h_min=ceiling(log_2(8+1)) = 4,h_min(5) = 3,h_min(12)=4
    a. H_max = floor(1.44*$\log_2$(n+2)-.328) -> 5,4,6
c. 2-4-Tree h_min=ceiling(log_4(8+1)) = 2,h_min(5) = 2,h_min(12)=2
    a. H_max = log_2(n+1) = 3,3,4
d. Red-Black Tree h_min=ceiling(log_2(8+1)) = 4,h_min(5) = 3,h_min(12)=4
    a. H_max = Check the tree and give marks if they are close enough with one path having alternate red and black nodes.

Problem 1:new tree with more readability.

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